

BME 4810 – Final Project Report

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Overview

This project develops and evaluates machine learning pipelines for automated four-class classification of retinal OCT images from the OCT-MNIST benchmark, using both classical sklearn ensemble models and deep learning architectures (CNN, MLP, RNN) to identify Normal retina and three disease states: Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen.

Methods

The project follows a two-track pipeline. The first track uses handcrafted features (HOG, Sobel edges, statistical descriptors, LBP textures) combined with scikit-learn ensemble classifiers. The second track uses TensorFlow/Keras neural network architectures operating directly on image pixels or feature representations. Both tracks use balanced stratified sampling to address class imbalance, 5-fold cross-validation for hyperparameter tuning, and evaluate final models on the held-out 1,000-image test set.

a. Preprocessing, Dimensionality Reduction, or Feature Generation

The OCT-MNIST training set contains 97,477 images across four highly imbalanced classes: Normal (33,484), Drusen (46,026), CNV (10,213), and DME (7,754). Training on the raw distribution would bias classifiers toward Normal and Drusen. To address this, balanced stratified sampling was applied: 5,000 images per class (20,000 total) for sklearn models and 2,436 per class (9,744 total) for neural networks, both using a fixed random seed for reproducibility.

Milestone 1 evaluated four filtering strategies: no filter, Gaussian, Median, and Sobel applied to images downsampled to 14×14. Sobel edge filtering achieved the highest Random Forest cross-validation accuracy (65.1%), capturing retinal layer boundary gradients that are the primary discriminative structure across all four classes. This established Sobel as the foundation for all subsequent feature extraction.

The final sklearn feature set, `F_rich_v6`, concatenates five components: HOG features (144 dims, 9 orientation bins, 8×8 cells, L2-Hys normalization) capturing directional gradient structure; raw pixel values (784 dims); Sobel-filtered and max-pooled edge maps (196 dims); statistical descriptors computed on the Sobel images (18 dims) including mean, standard deviation, quartiles, quadrant means, gradient variance, energy, and entropy; and Local Binary Pattern texture histograms (18 dims, P=16, R=2, uniform mapping) targeting micro-texture differences between DME fluid cysts and Drusen deposits. The total feature dimensionality is 1,160. After `StandardScaler` normalization, PCA with 99% variance retention reduced the space to 353 components. For neural networks, a 100-component PCA projection was used for the MLP input, while the CNN and RNN operated directly on raw 28×28-pixel arrays.

b. Preliminary Designs

Sklearn Preliminary Designs (Milestone 2)

Three preliminary sklearn classifiers were evaluated on F_{rich} (998 features) with PCA 95% reduction (~94 components) and 5-fold stratified cross-validation on the balanced training set:

Design 1: SVM with RBF Kernel: Trained on PCA-reduced F_{rich} features. The RBF kernel captures non-linear class boundaries in the feature space. Hyperparameters $C \in \{0.1, 1, 10\}$ and $n_estimators \in \{10, 20\}$ were explored with Bagging. Best test accuracy: 60.6%.

Design 2: RF $\rightarrow$ Logistic Regression: A Random Forest (200 trees) ranked features by importance on F_{rich} , selecting the top 90% cumulative importance features (796 features). Logistic Regression was then trained on this reduced set with $C \in \{0.1, 1, 10, 50\}$. This pipeline combines non-linear feature selection with a linear classifier. Best test accuracy: 44.3%.

Design 3: KNN: K-Nearest Neighbors on PCA-reduced features with $k \in \{3, 5, 7, 9, 11, 15\}$. Provides a non-parametric baseline that measures class separability in PCA feature space. Best test accuracy: 48.8%.

PCA was confirmed as the superior dimensionality reduction strategy over LDA, which is restricted to $n_classes - 1 = 3$ components and discards 99.7% of variance. The SVM design was selected as the strongest base for further development.

Neural Network Preliminary Designs (Milestone 3)

Two neural network architectures were designed as preliminary models. I designed and implemented the CNN and RNN; my teammate Srihitha Mitta improved the MLP architecture.

Design 1: CNN (Convolutional Neural Network): I designed a two-block convolutional architecture. Each block has two Conv2D layers (3×3, same padding) with BatchNormalization, ReLU activation, and MaxPooling2D (2×2). Block 1 uses 32 filters; Block 2 uses 64 filters, reducing spatial dimensions from 28×28 to 7×7. The classification head is: Flatten (3,136) $\rightarrow$ Dense (256, ReLU) $\rightarrow$ Dropout (0.5) $\rightarrow$ Dense (4, softmax). BatchNormalization stabilizes training and Dropout prevents overfitting.

Design 2: RNN (Bidirectional LSTM): I designed the RNN to treat each OCT image as a temporal sequence of 28 row vectors. A Bidirectional LSTM layer processes the sequence in both directions, capturing both top-to-bottom and bottom-to-top retinal layer dependencies. This tests whether sequential spatial structure across image rows is exploitable for OCT classification.

c. Final Designs and Fine-Tuning

Sklearn Final Design

The final sklearn pipeline uses F_{rich_v6} (1,160 features) with StandardScaler and PCA 99% (353 components) on a balanced set of 20,000 images. Three ensemble strategies were implemented, all tuned via 5-fold cross-validation. For each model component, hyperparameter grids were explored: SVM with $C \in \{1, 5, 10, 20, 50\}$ (best: $C=20$, CV acc=75.6%), ExtraTreesClassifier with $n_estimators \in \{100, 200, 500\}$ (best: 500, CV acc=64.6%), and HistGradientBoostingClassifier with $learning_rate \in \{0.05, 0.1, 0.2\}$ (best: 0.1, CV acc=67.9%).

Bagging SVMs: 15 SVC estimators ($C=20$, RBF, probability=True), each trained on 70% bootstrap subsamples, with probability averaging across estimators.

Soft Voting (SVM + ExtraTrees + HGB): Three classifiers with distinct inductive biases combined by averaging class probability outputs. The diversity of margin-based, tree-based, and gradient-boosting predictions reduces ensemble variance.

Stacking (SVM + ExtraTrees + HGB $\rightarrow$ LR): Out-of-fold probability predictions from the three base classifiers are fed to a Logistic Regression meta-learner trained to optimally blend the base outputs.

Neural Network Final Design

The CNN was selected as the final neural network model based on its superior test accuracy (76.9%) compared to the RNN (66.7%) and improved MLP (64.6%). All models used: Adam optimizer ($\text{lr}=1\times 10^{-3}$, weight decay= 1×10^{-4}), Sparse Categorical Cross-Entropy loss, batch size 64, 80/20 stratified train/validation split (7,795/1,949 images), with ReduceLROnPlateau (factor=0.5, patience=5) and EarlyStopping (patience=10, restore_best_weights=True) callbacks.

Results

Sklearn Model Results

Figure 1 shows the accuracy and macro F1 comparison across the three sklearn ensemble models. Soft Voting achieves the highest test accuracy (72.3%) and macro F1 (70.6%), representing an 8.5 percentage point improvement over the Milestone 3 baseline (63.8%). All three models exceed the baseline.

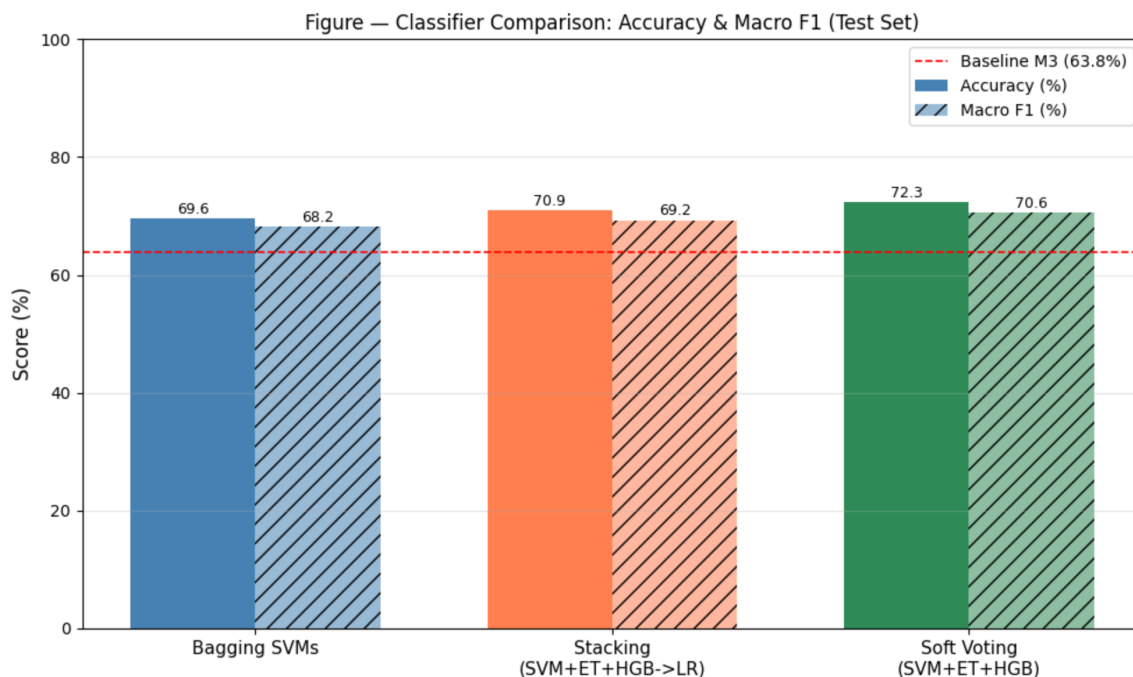


Figure 1. Accuracy and Macro F1 comparison for the three sklearn ensemble models on the test set ($n=1,000$). The red dashed line shows the Milestone 3 baseline (63.8%).

Table 1 summarizes the complete quantitative metrics for all three models.

Model	Accuracy	Precision	Recall	F1	AUC
Bagging SVMs	0.6960	0.6924	0.6960	0.6818	0.8998
Stacking (SVM+ET+HGB→LR)	0.7090	0.7115	0.7090	0.6922	0.9007
Soft Voting (SVM+ET+HGB)	0.7230	0.7245	0.7230	0.7059	0.9111
Baseline M3 Stacking	0.6380	—	—	—	—

Table 1. Sklearn ensemble results (macro-averaged metrics). All values from the 1,000-image test set.

Figure 2 shows the row-normalized confusion matrices (recall per class) for all three models. DME consistently shows the lowest recall across all models (0.38), indicating systematic misclassification as Normal or Drusen. CNV achieves the highest recall (0.83–0.85), as choroidal neovascularization produces distinctive sub-RPE lesion patterns. Normal and Drusen both achieve strong recall (≥ 0.76 and ≥ 0.81 respectively).

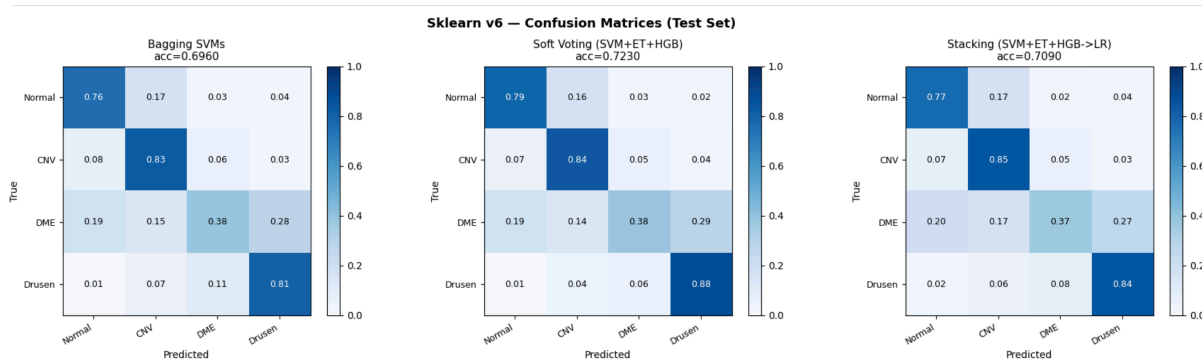


Figure 2. Row-normalized confusion matrices for Bagging SVMs (acc=0.696), Soft Voting (acc=0.723), and Stacking (acc=0.709) on the test set. Values represent recall per class.

Figure 3 presents the ROC curves for all three models (macro-averaged, left) and per-class ROC curves for the best model, Soft Voting (right). The macro-averaged AUC values are 0.900 (Bagging), 0.901 (Stacking), and 0.911 (Soft Voting). Per-class AUC confirms that Drusen (0.943) and Normal (0.938) are the most separable classes, while DME (0.842) remains the most challenging due to its overlap with both Normal and Drusen feature distributions.

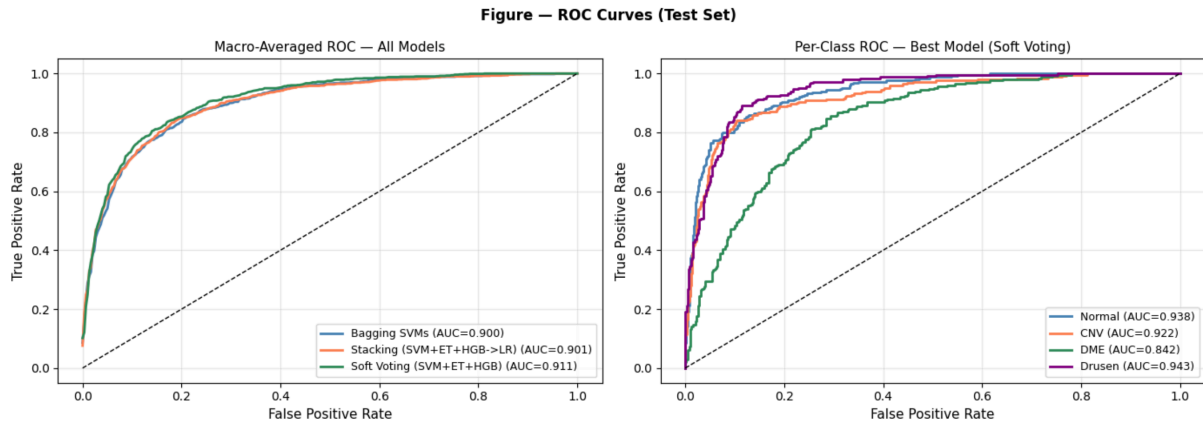


Figure 3. ROC curves on the test set. Left: macro-averaged ROC for all three sklearn models. Right: per-class ROC for the best model (Soft Voting, macro AUC=0.911).

Neural Network Results

CNN

The CNN achieved 76.9% test accuracy after training for 40 epochs. Figure 4 shows the training and validation learning curves. Both loss curves converge smoothly after an initial instability at epoch 4 caused by a learning rate schedule adjustment. Training and validation accuracy stabilize around 90% and 90% respectively, indicating good generalization without significant overfitting.

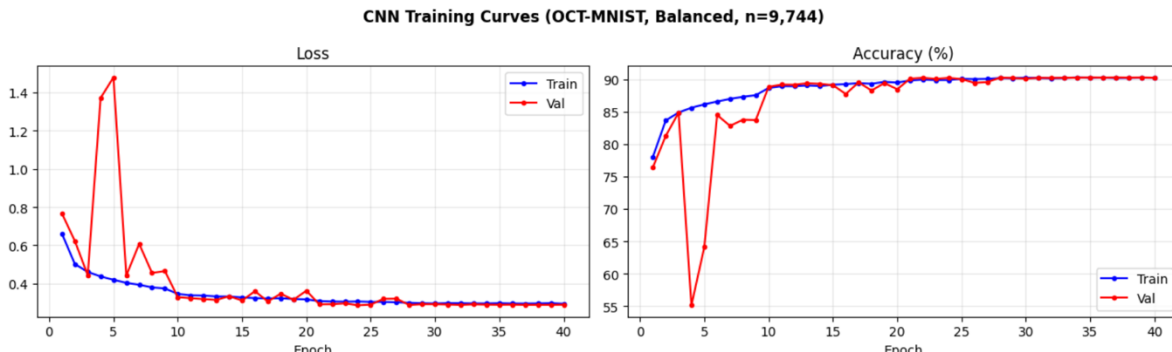


Figure 4. CNN training and validation loss (left) and accuracy (right) curves over 40 epochs.

Figure 5 shows the CNN confusion matrix. Normal class achieves near-perfect recall (0.97) due to the absence of pathological structures. CNV recall is strong (0.80) as choroidal lesions produce distinctive convolutional filter responses. DME recall is only 0.35 despite high precision (0.96), reflecting systematic under-detection. Drusen achieves the best-balanced performance (precision 0.75, recall 0.95).

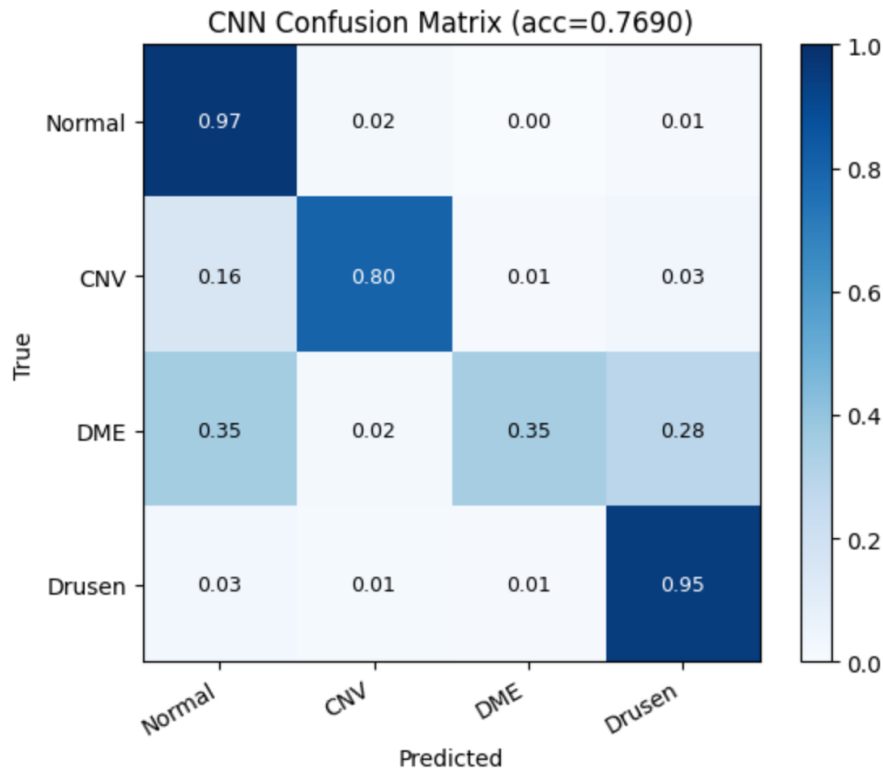


Figure 5. CNN confusion matrix (acc=0.769, test set n=1,000). Row-normalized; values represent recall per class.

MLP (Improved by Srihitha Mitta)

The improved MLP achieved 64.6% test accuracy. Figure 6 shows its learning curves: the divergence between training (82%) and validation accuracy (69%) at epoch 40 indicates overfitting, as the MLP operates on 314-dimensional F_{rich} PCA features without the spatial regularization provided by convolutional operations.

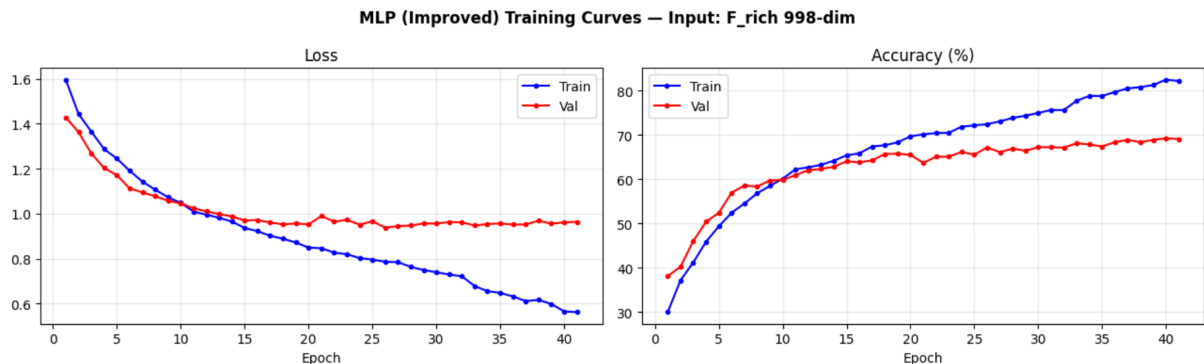


Figure 6. MLP training and validation curves over 40 epochs. The growing gap between training and validation accuracy after epoch 15 indicates overfitting.

The MLP confusion matrix (Figure 7) shows more balanced per-class performance than the CNN, with DME recall improved to 0.57, but at the cost of lower overall accuracy due to weaker spatial feature discrimination.

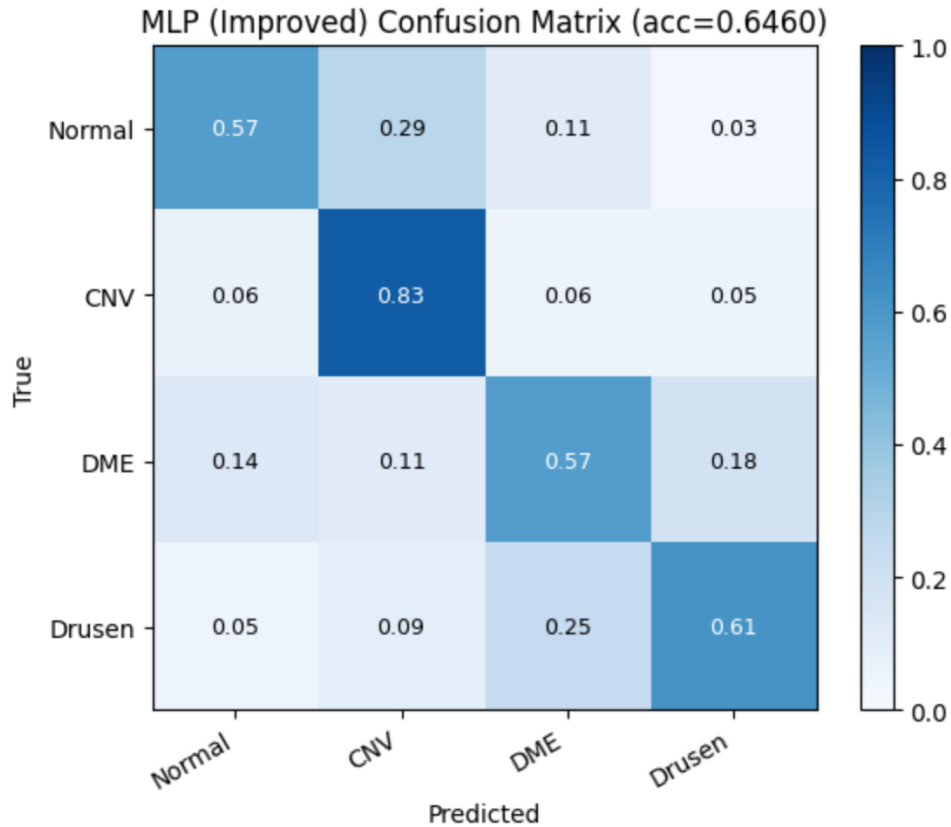


Figure 7. MLP confusion matrix (acc=0.646, test set). Improved by Srihitha Mitta.

RNN

The Bidirectional LSTM RNN achieved 66.7% test accuracy. Figure 8 shows high validation loss oscillations throughout training, reflecting the difficulty of learning stable sequential patterns from 28-pixel row sequences. The RNN treats each image row as a time step, but retinal layer patterns are not strictly sequential in the same sense as natural language, limiting the LSTM inductive bias advantage.

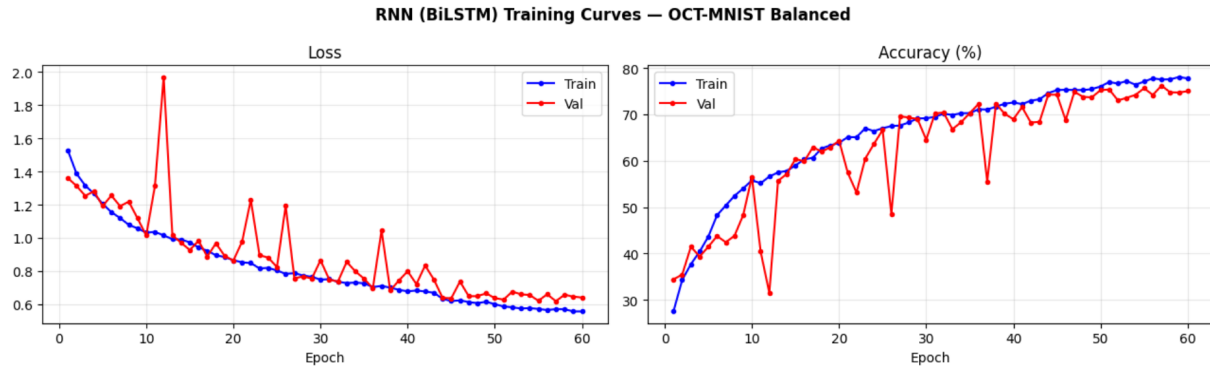


Figure 8. RNN (BiLSTM) training and validation curves over 60 epochs. High validation loss oscillation indicates unstable training dynamics.

Figure 9 shows the RNN confusion matrix. Compared to the CNN, the RNN shows markedly improved DME recall (0.52 vs. 0.35) but reduced Normal recall (0.76 vs. 0.97). This suggests the bidirectional LSTM captures some DME-specific vertical layer structure but loses the fine-grained local pattern recognition of CNNs.

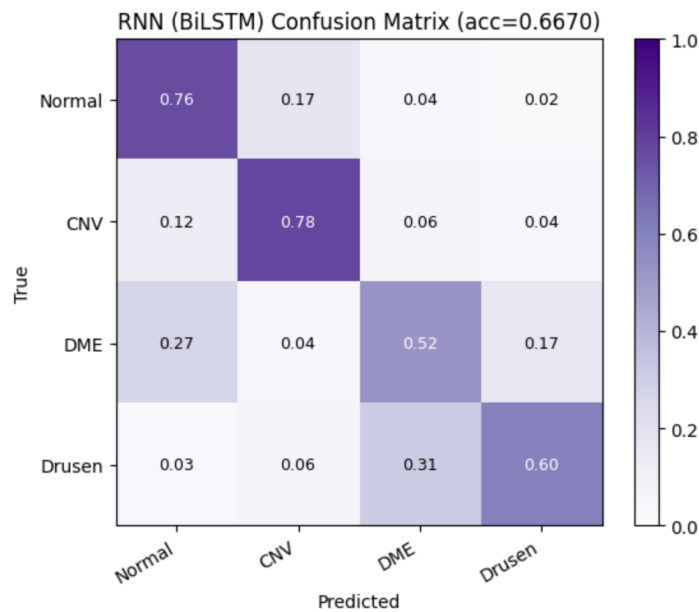


Figure 9. RNN (BiLSTM) confusion matrix (acc=0.667, test set).

Neural Network Comparison

Figure 10 overlays the training curves for all three neural network models, confirming the CNN's superior convergence. The CNN reaches 90% validation accuracy within 10 epochs, while MLP and RNN plateau near 69% and 75% respectively.

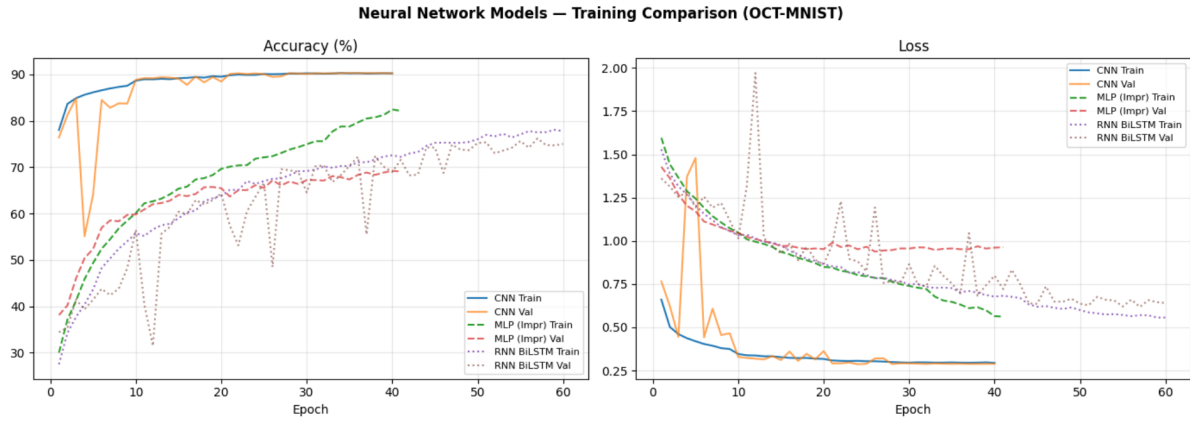


Figure 10. Training comparison across all three neural network models: CNN (solid), MLP Improved (dashed), RNN BiLSTM (dotted). Left: accuracy. Right: loss.

Cross-Track Comparison

Model	Type	Test Acc	Macro F1	Macro AUC
CNN	Deep Learning	0.7690	0.750*	—
Soft Voting (SVM+ET+HGB)	Sklearn Ensemble	0.7230	0.7059	0.9111
Stacking (SVM+ET+HGB→LR)	Sklearn Ensemble	0.7090	0.6922	0.9007
RNN (BiLSTM)	Deep Learning	0.6670	0.660*	—
Bagging SVMs	Sklearn Ensemble	0.6960	0.6818	0.8998
MLP (Improved)	Deep Learning	0.6460	0.640*	—
Baseline M3 Stacking	Sklearn	0.6380	—	—

Table 2. Complete model comparison. *F1 approximate from classification report. All models evaluated on the same 1,000-image test set.

Conclusions

This project successfully developed a complete machine learning pipeline for OCT retinal disease classification, achieving 76.9% test accuracy with the CNN and 72.3% with the best sklearn ensemble (Soft Voting), both substantially exceeding the Milestone 3 baseline of 63.8%. The CNN is the most promising model, exploiting spatial structure through learned convolutional filters that outperform handcrafted HOG and Sobel features. DME remains the most challenging class across all models due to the visual similarity of

fluid accumulation patterns with Normal retina and Drusen deposits at 28×28 resolution, suggesting that class-weighted training or higher-resolution inputs would be the most impactful next step.

Contributions

The following describes contributions across the project:

- Bhakti Moradiya: All Milestone 1 work (filtering strategies, feature generation, F_rich pipeline); sklearn pipeline development across Milestones 2 and 3 (balanced sampling, PCA/LDA analysis, tuning of SVM, RF→ LR, KNN); final sklearn ensemble development (HOG features, v6 pipeline, Soft Voting, Bagging, Stacking); CNN architecture design and implementation; RNN (BiLSTM) architecture design and implementation.
- Srihitha Mitta: MLP architecture improvements (BatchNormalization, Dropout tuning); contributions to Milestone 2 classifier comparison and RF feature importance analysis; joint authorship of progress reports. Results from Srihitha's MLP training (Figure 7) were provided as notebook outputs and figures only, per the project policy prohibiting sharing of code or written work.

Self-Reflection

The course effectively met its stated objectives of designing ML pipelines for real biomedical datasets and systematically comparing classical and deep learning approaches. Working through the progressive milestone structure made the tradeoffs between handcrafted and learned features concrete: it was not until adding HOG features in Milestone 4 that the sklearn models meaningfully closed the gap with neural networks, illustrating why feature engineering choices matter. The feedback loop through graded milestones, particularly the confusion matrix normalization correction and the SVM Bagging critique, were valuable in building habits of methodological rigor that I will carry forward.

Appendix

The following executed Jupyter notebooks are submitted alongside this report:

sklearn_models.ipynb — Final sklearn ensemble pipeline (F_rich_v6, Soft Voting, Bagging, Stacking, ROC curves, confusion matrices, bar chart).

NeuralNetwork_Models_trial2.ipynb — Neural network models (CNN, MLP Improved, RNN BiLSTM) with training curves, confusion matrices, and comparative analysis.